

Mitigation Cost Curve

SLIBRARY 12 · SHEET 5 / 5

Diminishing returns on de-risking

PROFESSIONAL DEFINITION

The Mitigation Cost Curve shows how the cost of reducing a risk rises while the marginal risk removed falls, revealing the point beyond which further de-risking costs more than the risk it removes.

WHO DEVELOPED IT

A risk-economics construct related to the economics of safety and diminishing returns to control.

WHY IT WAS DEVELOPED

It was developed to identify the optimal level of mitigation, avoiding both reckless under-investment and uneconomic over-control.

FIGURE 1 · MITIGATION COST CURVE

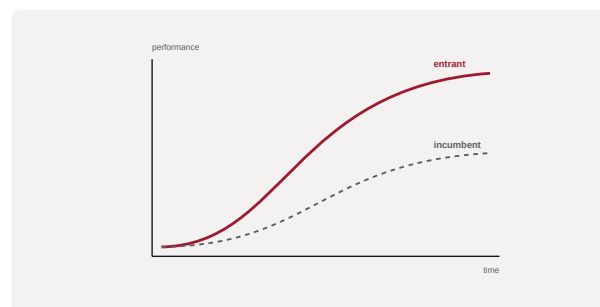


Figure 1. A schematic of the Mitigation Cost Curve as applied in BARBM312 (illustrative).

HOW IT IS USED

Mitigation cost is plotted against residual risk; investment stops where marginal cost exceeds marginal risk reduction.

WHEN IT IS USED

When deciding how much to invest in AI risk controls.

BY WHOM IT IS USED

Risk, finance and governance leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It introduces economic discipline to de-risking, recognising that perfect safety is neither attainable nor optimal.

RELATED FRAMEWORKS IN THIS COURSE

Cost-Benefit Trade-off Matrix · Slibrary 12

AI Risk Register · Slibrary 12

Bias & Failure Modes Map · Slibrary 12

Black Swan Radar · Slibrary 12

SOURCES (HARVARD REFERENCING STYLE)

Viscusi, W.K. (1996) 'Economic foundations of the current regulatory reform efforts', Journal of Economic Perspectives, 10(3).
Hubbard, D.W. (2020) The Failure of Risk Management. Hoboken: Wiley.